

EXOHUB MATRIX

ASTRONOMY CLUB, IITK

ABOUT THE PROJECT

The project aims to cover a wide domain of astronomy and its real world application of ML. The project starts from very basic and touches advanced tech aspect for **detection of habitable exoplanets** useful for future human endeavors.

Our aim is to cover exoplanet detection techniques, study **planetary properties** using observational data, use **spectroscopy** for atmosphere analysis and apply **ML** techniques to assess **exoplanet habitability**.

Altitude-Azimuth

1) The **altitude** is the angle above the horizon. The **Azimuth** is the angle measured along the horizon from the north point. This system is based on the local position of the observer.

Equatorial

2) **Equatorial** system projects the Earth's equator in the sky and uses Right Ascension and Declination. RA is similar to longitudes and DEC is similar to latitudes on earth.

Ecliptic

3) **Ecliptic coordinate system** is based on the ecliptic plane.

Galactic

4) **Galactic coordinate system** utilizes the plane of Milky Way galaxy as its fundamental reference plane.

Supergalactic

5) **Super Galactic Coordinate System** represents a coordinate framework on a large scale.

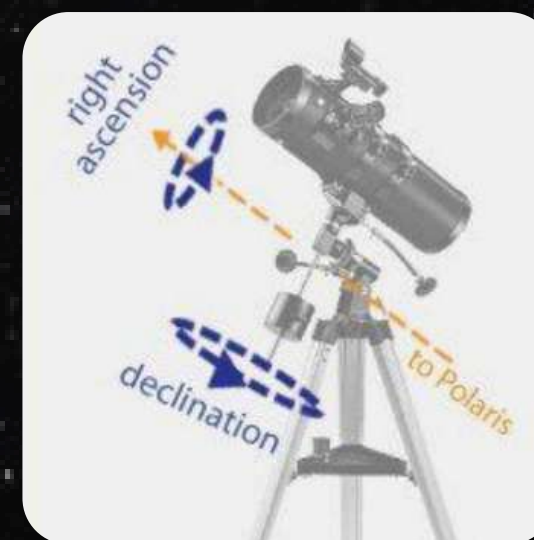
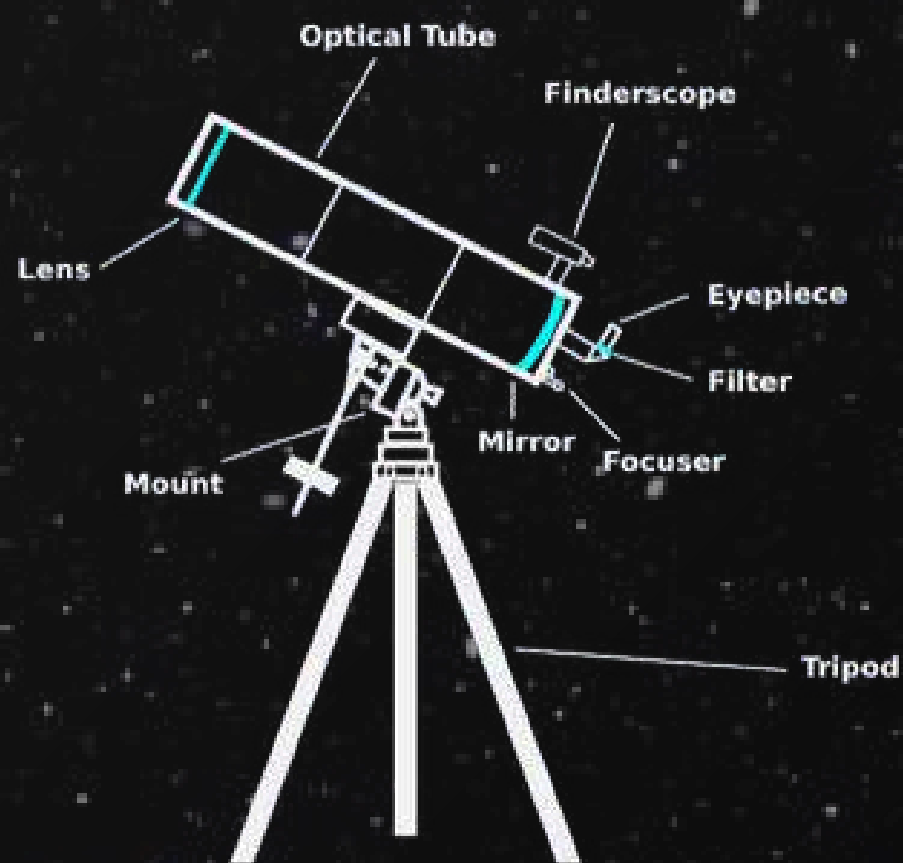
Telescope magnifies distant objects using curved mirrors and lenses, based on refraction and reflection principles.

Types of telescopes

Optical Design

Mount

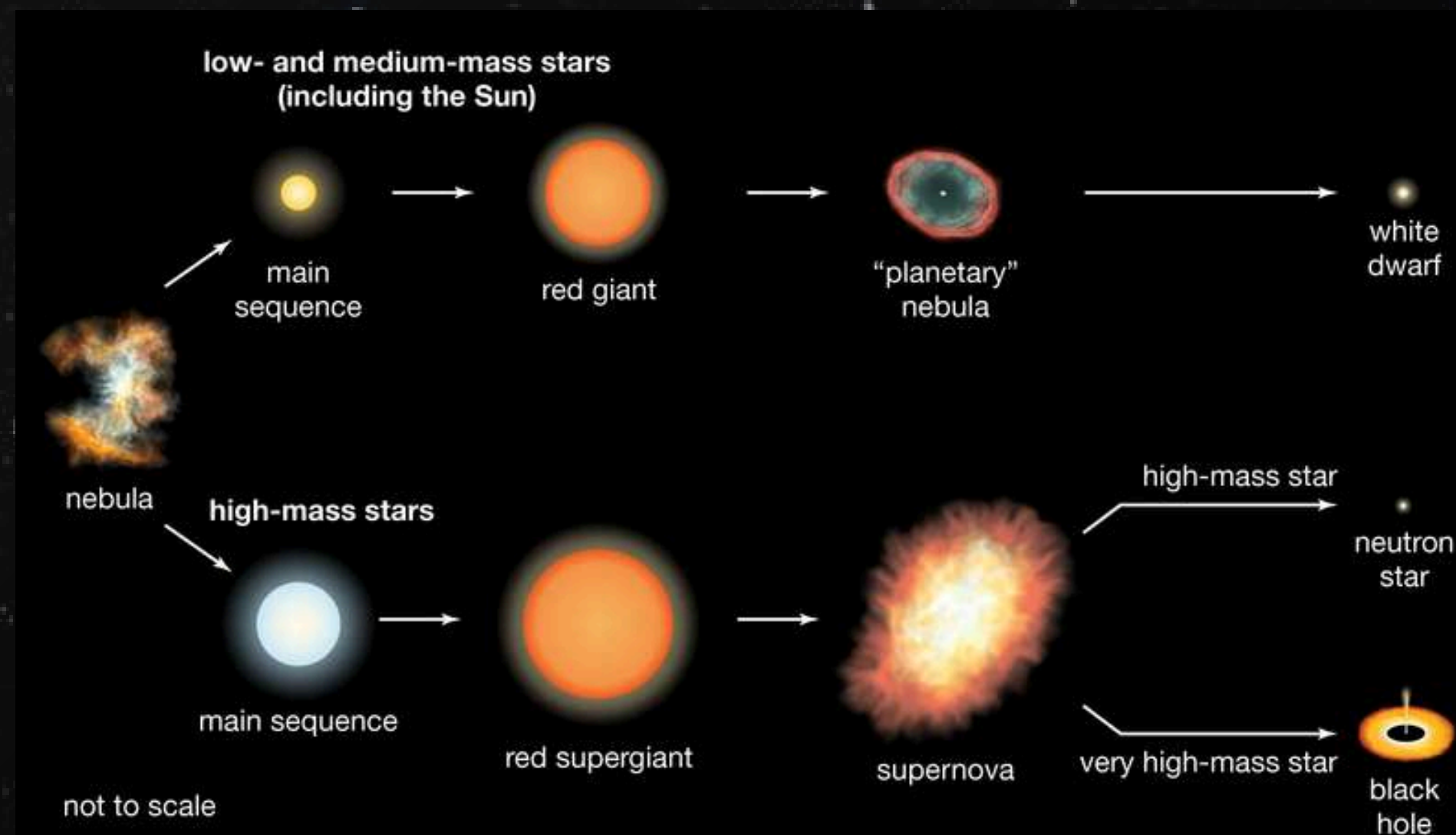
Parts of a telescope



Students captured this image during the project.

Every star begins its life as a **nebula**. Gaseous particles come close under the influence of their own gravity to form stars.

The **mass of star** thus forms determines the path it follows.



Black holes

A black hole is a dense region where gravity is so strong that nothing, not even light, can escape. It forms from the collapse of massive stars and is detected by its effect on nearby matter.

more than
— 20 times the mass
of the Sun

The fate of dying
star is determined
by it's mass

Neutron stars

A neutron star forms from the collapsed core of a massive star (8–20 times the Sun's mass) after it exhausts its fuel and explodes in a supernova. The core becomes incredibly dense, packed into a small, supermassive object.

8 to 20 times the
mass of the Sun

White Dwarf

A white dwarf is the dense, hot core left behind after a low- or medium-mass star exhausts its fuel and sheds its outer layers. It no longer undergoes fusion and gradually cools over billions of years.

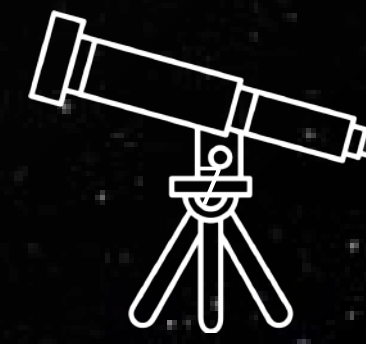
up to about 8
times the mass of
the Sun

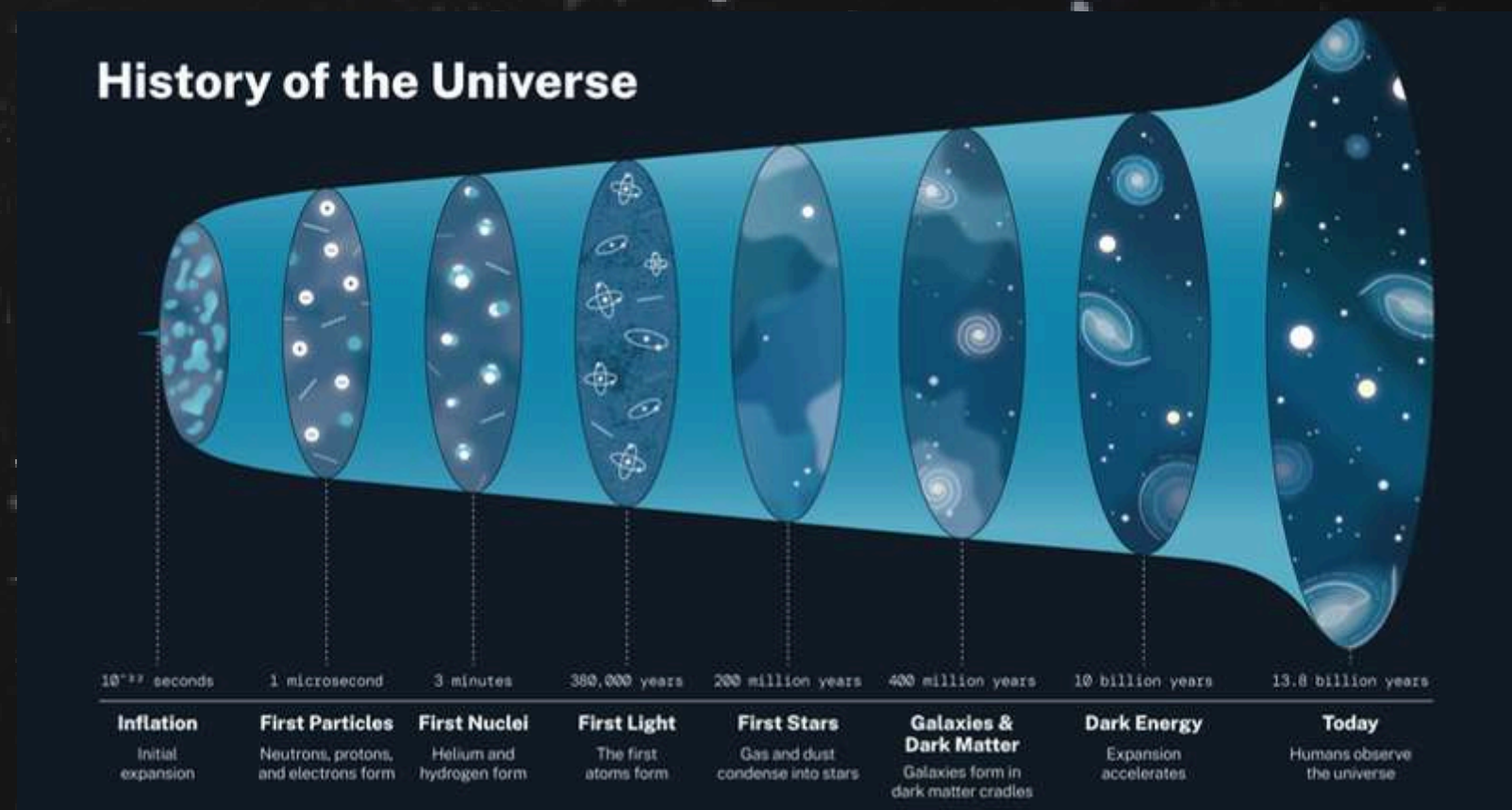
Three primary theories of the universe's evolution

**Big Bang
Theory**

**Steady State
Theory**

**Pulsating
Universe Theory**

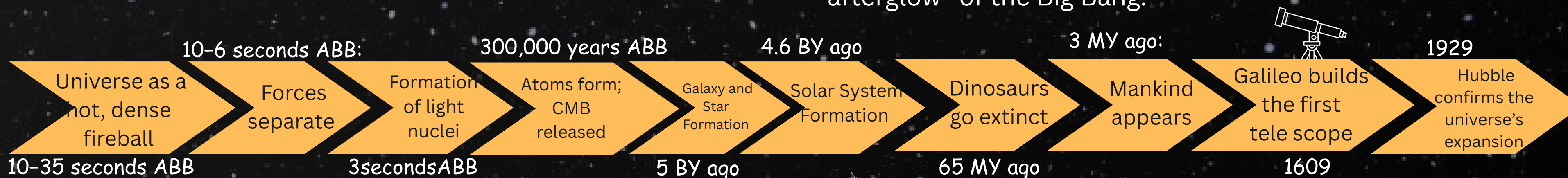




At the beginning of the universe, all matter was concentrated in an extremely hot and dense fireball ($\sim 10^{12}$ K). Around 14 billion years ago, a massive explosion—known as the Big Bang—marked the beginning of space and time.

Key Observational Evidence:

- **Redshift of Galaxies:** Indicates that the universe is expanding.
- **Cosmic Microwave Background (CMB):** Remnant heat from the early universe, acting as an "afterglow" of the Big Bang.

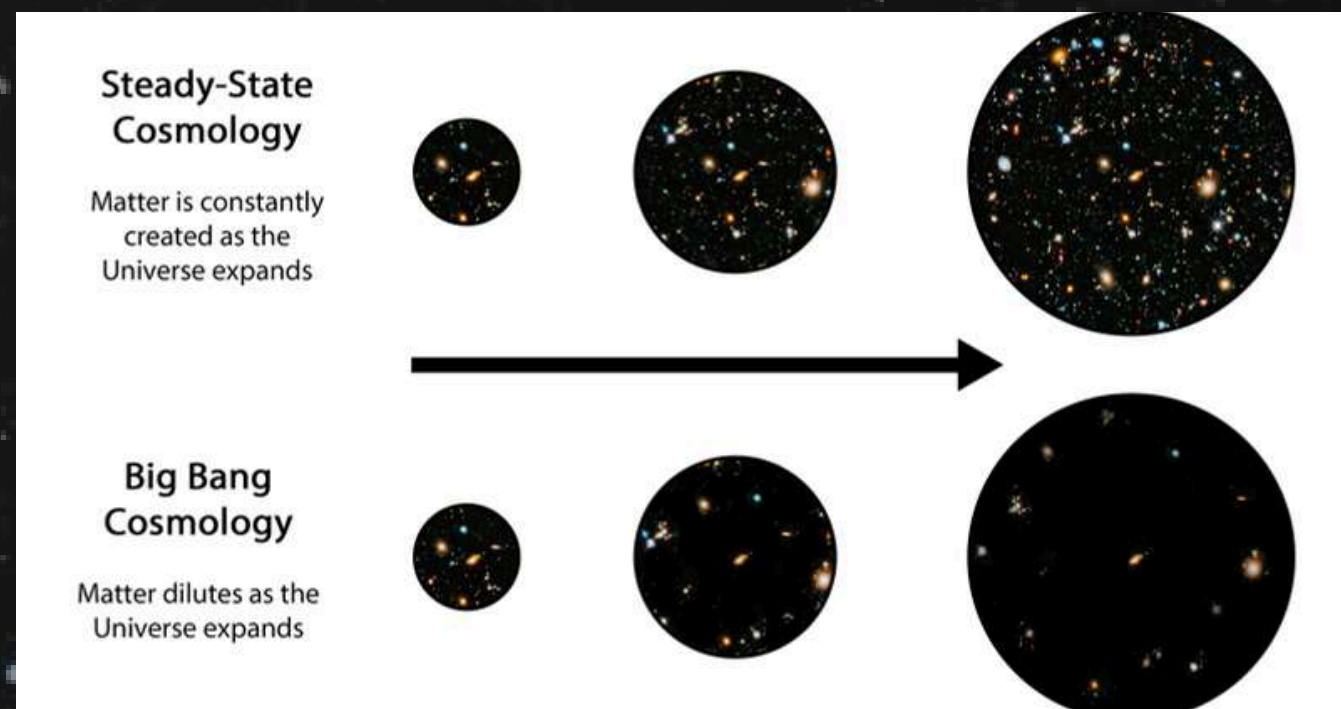


Galaxy and Star Formation

Solar System Formation

THE STEADY STATE THEORY

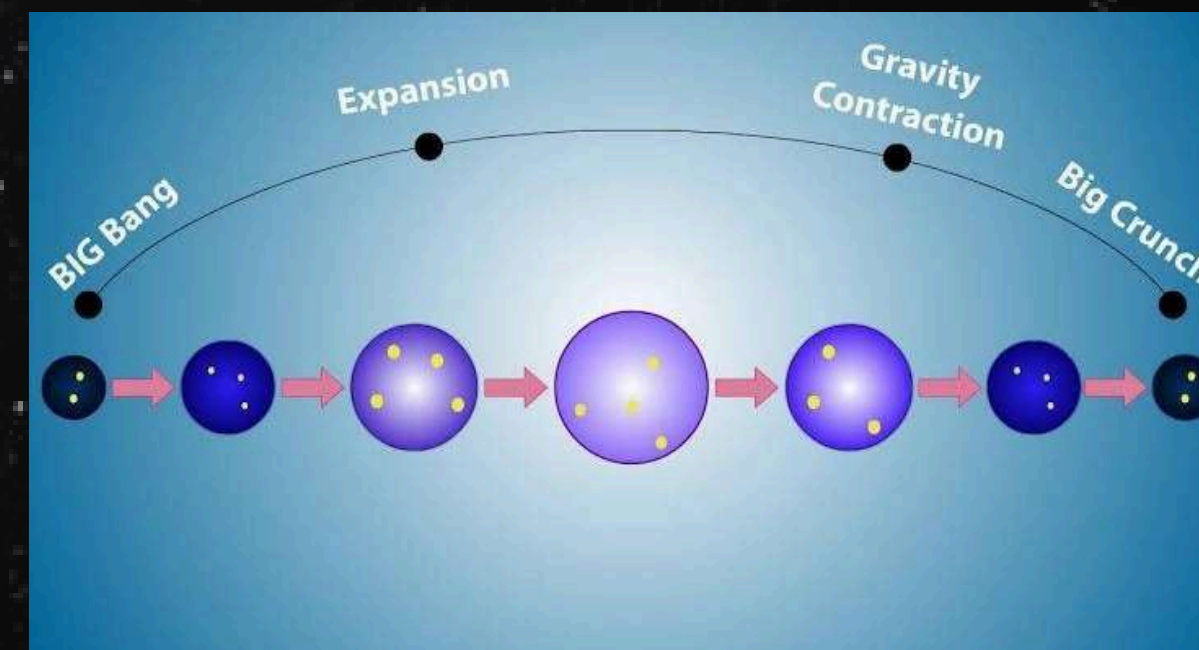
- Theory proposes a universe that is uniform in space and time, with no beginning or end. To maintain constant density despite expansion, new matter is continuously created to fill gaps left by receding galaxies



- The universe is uniform, eternal, and unchanging, with new matter continuously created to keep its density constant.
- Observations show that the universe has changed over time. The discovery of cosmic microwave background radiation supports the idea that it began as a hot, dense state—refuting the steady state theory of constant matter creation.

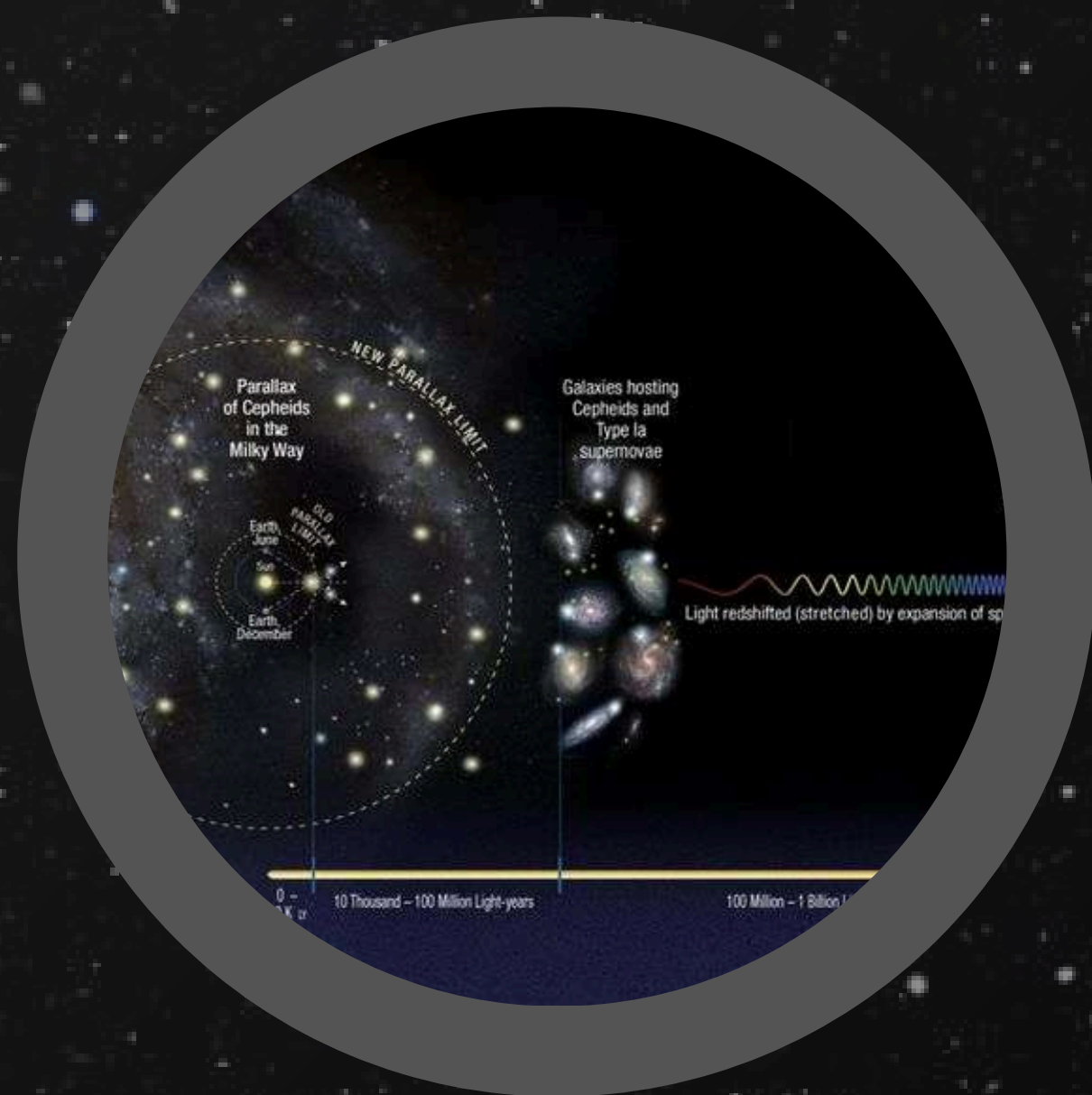
THE PULSATING UNIVERSE THEORY

- According to this theory, the universe is supposed to be expanding and contracting alternately i.e. pulsating.



- The universe will eventually stop expanding, contract due to gravity, collapse in a "Big Crunch," and possibly restart with another Big Bang.
- This cycle will continue forever, i.e., the universe will be oscillating or "pulsating" between the two events - Big Bang and Big Crunch.
- While the Pulsating Theory offers an intriguing alternative to a singular origin, it lacks substantial observational evidence. The Pulsating Theory remains a hypothesis

INTRODUCTION



The Cosmic Distance Ladder is a set of methods by which astronomers determine the distances to celestial objects using indirect techniques which involve mathematical concepts.

COSMIC DISTANCE LADDER

THE FIRST RUNG : EARTH'S RADIUS

Eratosthenes estimated Earth's radius using shadows, accurate within 8% of today's value.

THE SECOND RUNG : DISTANCE OF SUN

Aristarchus used the Moon's size and geometry to show the Sun is much larger than Earth.

THE THIRD RUNG : DISTANCE OF PLANETS

Copernicus used Mars' position and geometry to estimate its distance in the Solar System.

THE FOURTH RUNG : SPEED OF LIGHT

Huygens estimated speed of light to be 220,000km/s using Romer's observations

THE FIFTH RUNG : PARALLAX METHOD

Using angular shifts as observed from earth to measure distances to nearby stars

THE SIXTH RUNG : HR DIAGRAM

Comparing star's colours to it's absolute brightness to estimate distance

THE SEVENTH RUNG : CEPHEID VARIABLE STARS

This method can measure distances of galaxies upto 13 million light years away

THE EIGHTH RUNG : HUBBLE'S LAW

Hubble's Law uses redshift data and speed of light to measure distances across observable universe.

ASSIGNMENT - 1 : THEORETICAL ASTRONOMY

As part of our ongoing project, eight groups of 2–3 mentees each prepared and delivered brief 3–4 minute presentations on key topics in theoretical astronomy.

Carina Nebula



The Carina Nebula, about 7,500 light-years away, is a massive star-forming region hosting Eta Carinae and rich stellar clusters. It features dramatic structures like Mystic Mountain, showcasing active star birth and stellar feedback.

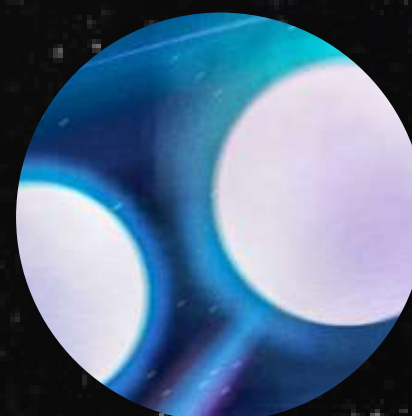
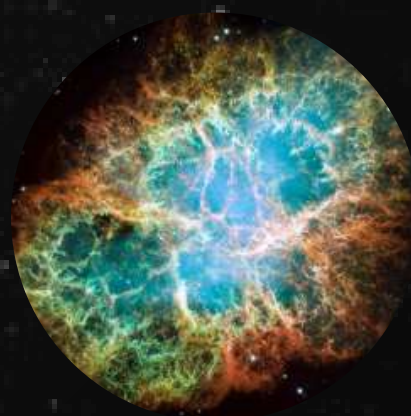
Heliocentrism



Proposed by Copernicus in 1543, the heliocentric model placed the Sun at the center with planets orbiting in circles. It explained retrograde motion and laid the foundation for modern astronomy and the Scientific Revolution.

Crab Pulsar

The Crab Pulsar is a fast-spinning neutron star powering the Crab Nebula. It emits across the spectrum and aids study of magnetospheres and particle acceleration.



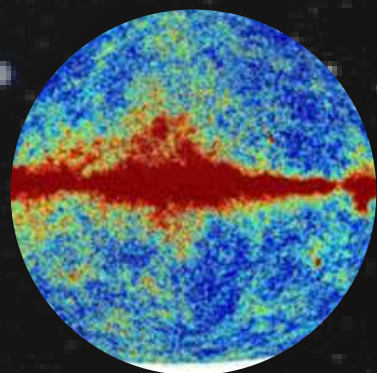
Binary Pulsars

Binary pulsars are systems with a spinning neutron star and a companion, often used to test general relativity. The Hulse–Taylor Pulsar, discovered in 1974, provided the first indirect evidence of gravitational waves.

ASSIGNMENT - 1 : THEORETICAL ASTRONOMY

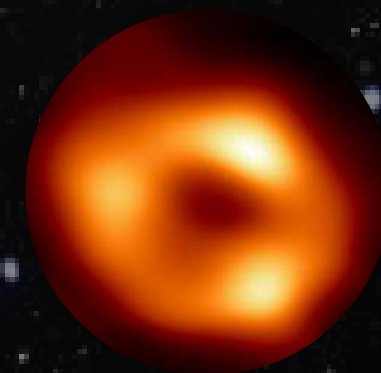
As part of our ongoing project, eight groups of 2–3 mentees each prepared and delivered brief 3–4 minute presentations on key topics in theoretical astronomy.

CMB



The Cosmic Microwave Background is ancient radiation from ~380,000 years after the Big Bang, mapped by COBE, WMAP, and Planck. Its tiny temperature fluctuations confirm inflation and early universe structure formation.

Sagittarius A*



Sagittarius A* is a supermassive black hole at the Milky Way's center, confirmed by stellar orbits like S2. It's studied via X-ray accretion, gas dynamics, and observations by EHT and JWST.

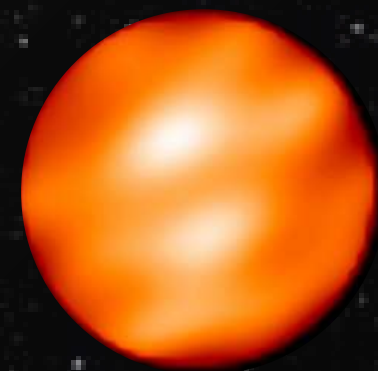
Sirius B

Sirius B, the nearest known white dwarf, is a dense, Earth-sized remnant with solar mass. Composed of carbon and oxygen, it confirms quantum and relativistic effects and helps us understand stellar death.



Betelgeuse

Betelgeuse, a red supergiant ~640 light-years away in Orion, is nearing the end of its life. Known for its 2019 dimming, it's expected to go supernova, offering insights into massive star evolution.



Our mentors organized the mentees into trios for an engaging quiz competition.

The Quiz comprised of the following topics :

- Astronomy 101
- Telescope Handling
- Universe Evolution
- Deep Sky Objects
- Drake Equation
- Celestial bodies formation
- General Astronomy
- Cepheid variables/cosmic distance ladder
- Exoplanet detection methods

01

TRANSIT METHOD

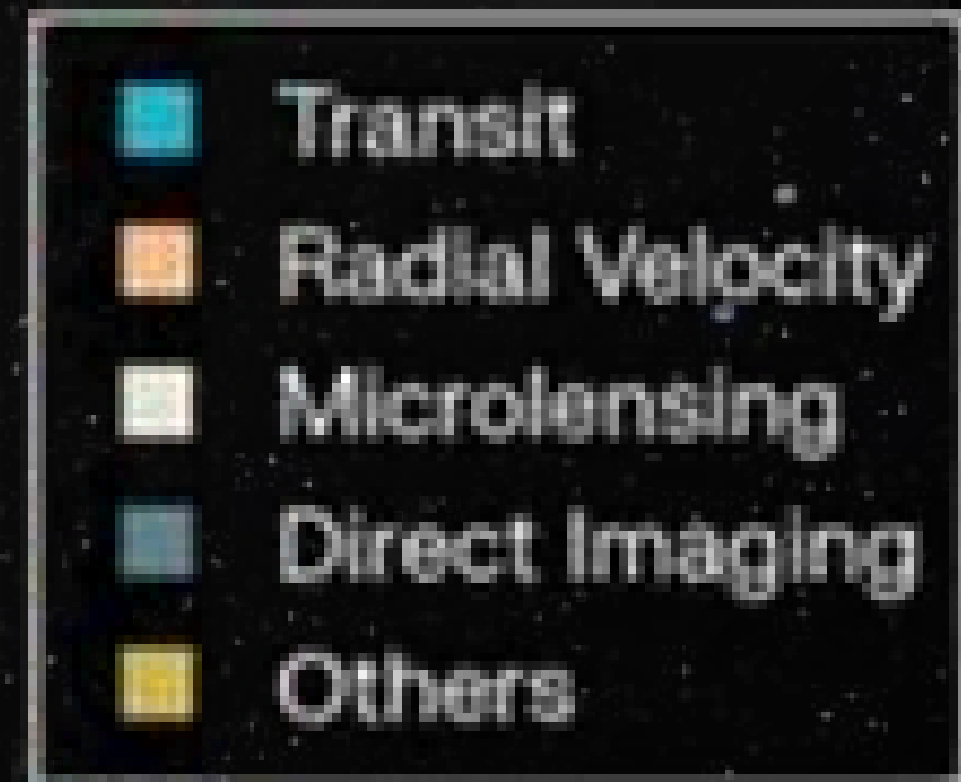
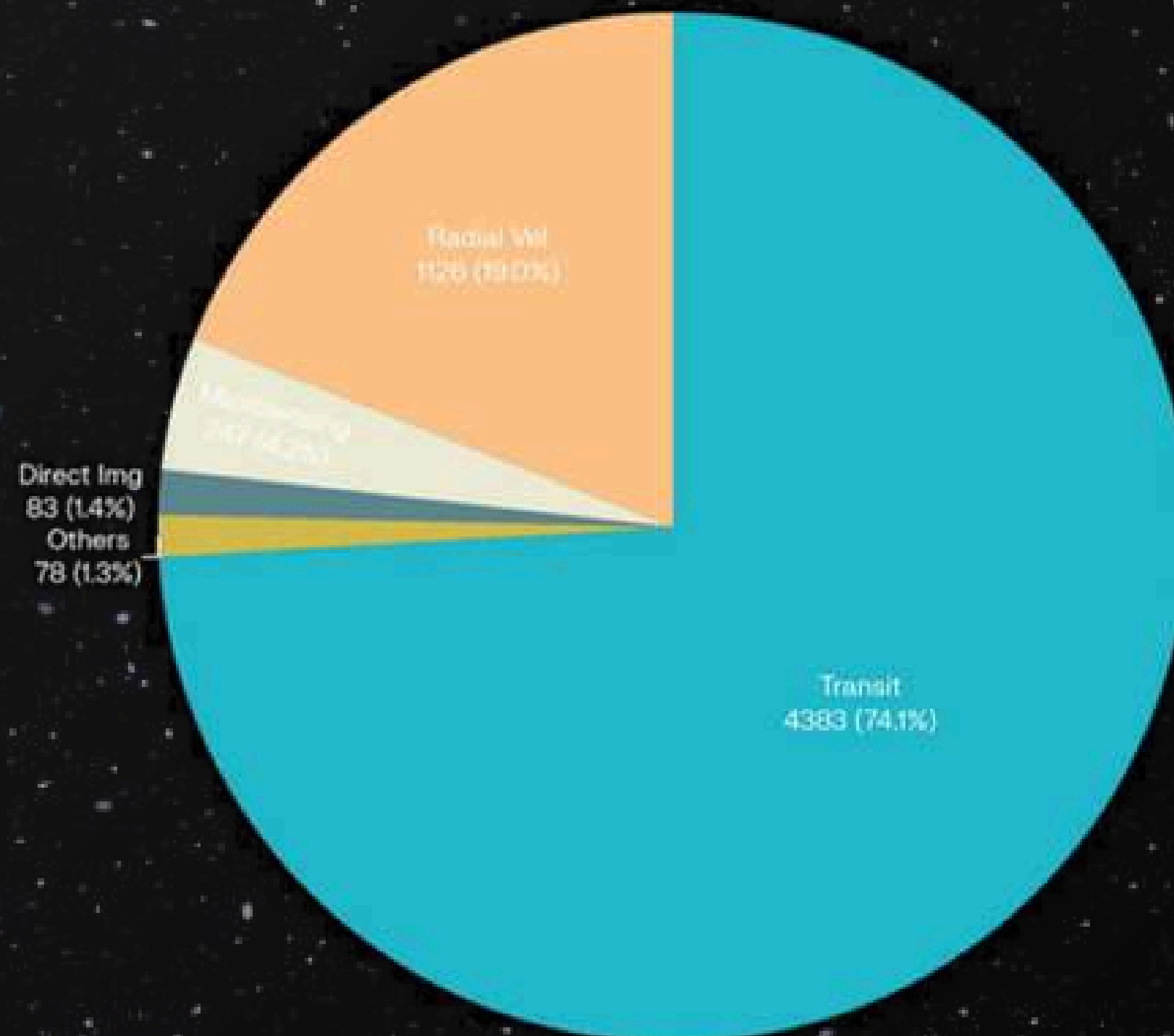
02

RADIAL VELOCITY

03

GRAVITY
MICROLENSING
EFFECT

04

DIRECT
IMAGING

Total: 5,917 Confirmed Exoplanets (as of June 2025)

01

TRANSIT METHOD

- The transit method detects exoplanets by measuring small dips in a star's brightness when a planet passes in front of it.
- Confirms planets and their size found by other methods.
- Hints at a planet's atmosphere through secondary dips in light.

02

RADIAL VELOCITY

- Detects planets by observing tiny shifts in a star's light due to its wobble from a planet's gravity but similar effects also shown by binary stars.
- This shift, seen through spectroscopy, is known as the Doppler Effect—light shifts blue when the star approaches and red when it moves away.

03

GRAVITY MICROLENSING EFFECT

Gravitational microlensing detects exoplanets when a star (and possibly its planet) passes in front of a more distant star, bending and focusing its light due to gravity.

03

Direct Imaging

Direct imaging captures light from an exoplanet when it's large, far from its star, and Adaptive optics and coronagraphs block the star's bright light, making the dim planet visible.

A FITS file (Flexible Image Transport System) is the standard format used in astronomy. It stores scientific data along with detailed metadata for analysis and interpretation. Universally adopted, it ensures long-term accessibility and cross-platform compatibility.



Applications

Images: Deep-sky surveys, galaxy structure, exoplanet detection.

Spectra: Elemental composition, redshift, temperature of stars.

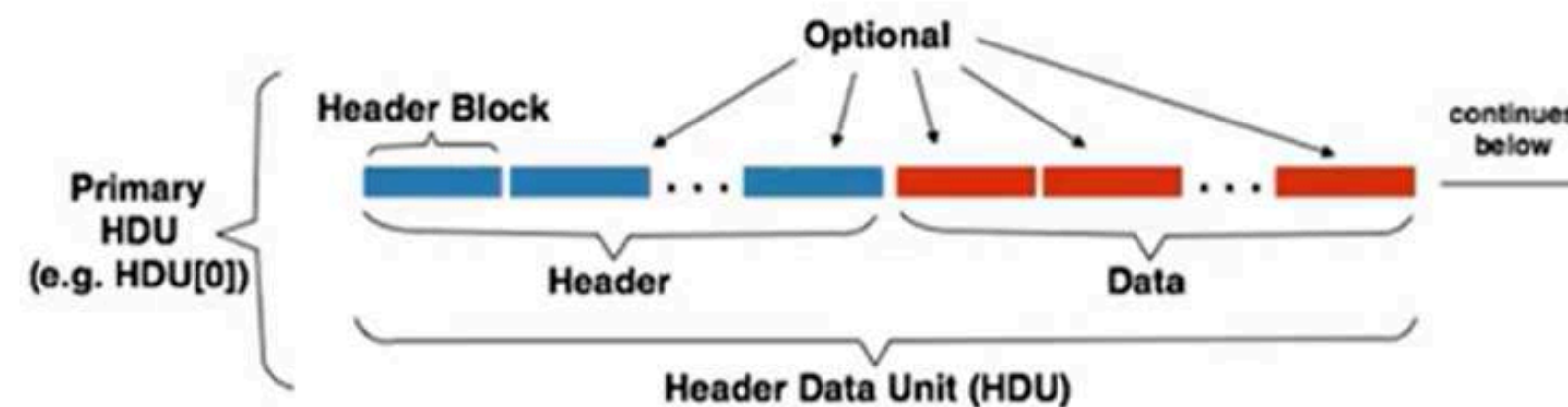
Time-Series: Light curves from variable stars, pulsars.

Tables: Catalogs of objects, calibration data.

Tools for Working with FITS

- Astropy.io.fits (Python)
- DS9 (visualization)
- FITS Liberator (for public image processing)
- The STScI Digitized Sky Survey

Structure of FITS File



Python Code to Read a FITS Image

```
from astropy.io import fits
f_name = 'ngc1261.fits'
hdulist = fits.open(f_name)
hdulist.info()
```

Filename: ngc1261.fits

No.	Name	Ver	Type	Cards	Dimensions	Format
0	PRIMARY	1	PrimaryHDU	161	(891, 893)	int16
1	xs.mask	1	TableHDU	25	1600R x 4C	[F6.2, F6.2, F6.2, F6.2]

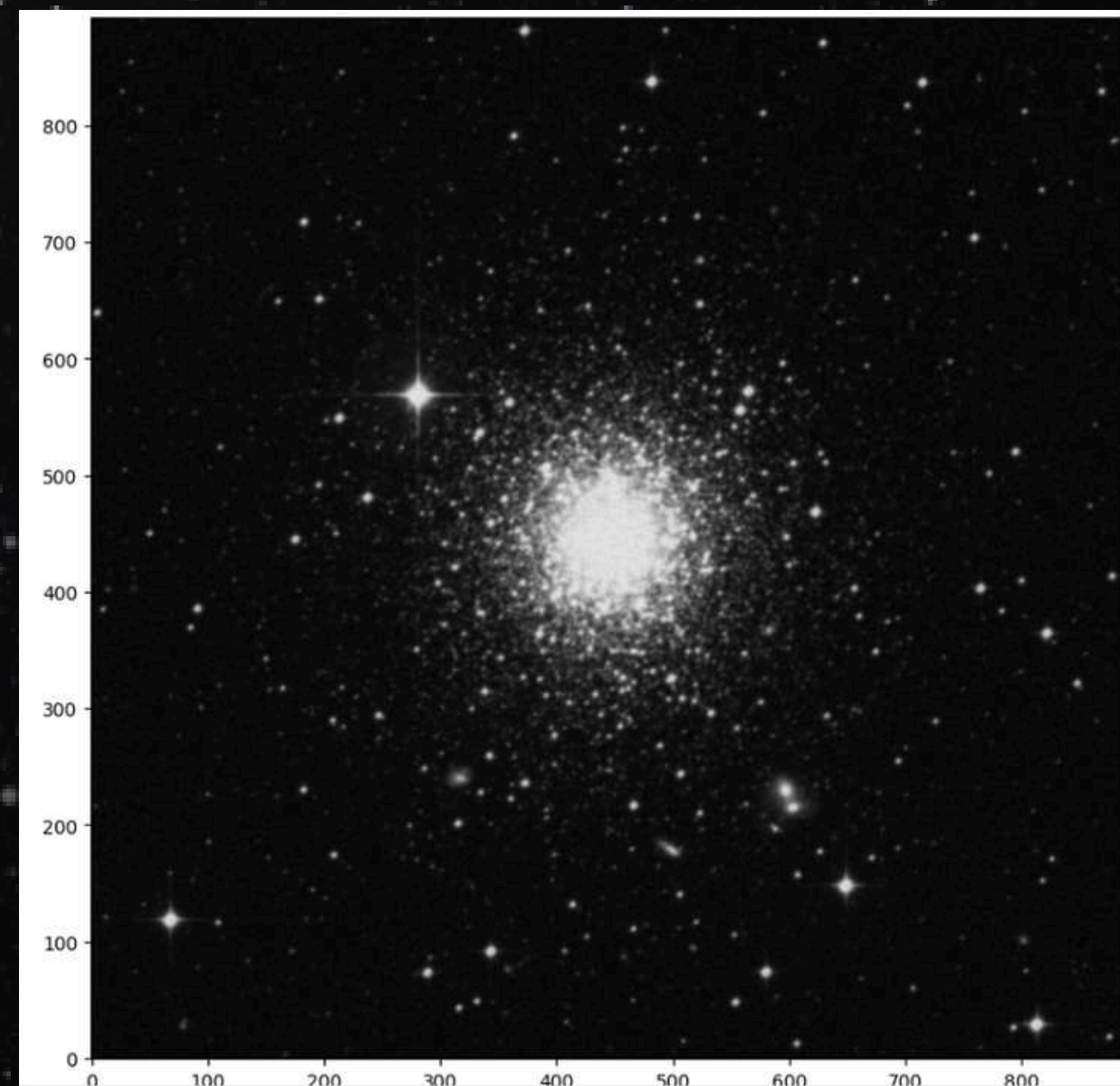
```
image=hdulist[0].data
print(image)
print(image.shape)
```

```
[[4444 4418 4630 ... 4430 3930 4143]
 [4444 4418 4205 ... 4217 3930 4143]
 [4444 4258 4258 ... 3978 4488 4701]
 ...
 [4214 3949 3949 ... 4789 4423 4423]
 [4002 3949 4374 ... 4576 4423 4423]
 [4002 4055 4480 ... 4576 4635 4423]]
(893, 891)
```

```
hdulist[0].header
```

```
SIMPLE = T /FITS: Compliance
BITPIX = 16 /FITS: I*2 Data
NAXIS = 2 /FITS: 2-D Image Data
NAXIS1 = 891 /FITS: X Dimension
NAXIS2 = 893 /FITS: Y Dimension
EXTEND = T /FITS: File can contain extensions
DATE = '2025-06-13' /FITS: Creation Date
ORIGIN = 'STScI/MAST' /GSSS: STScI Digitized Sky Survey
```

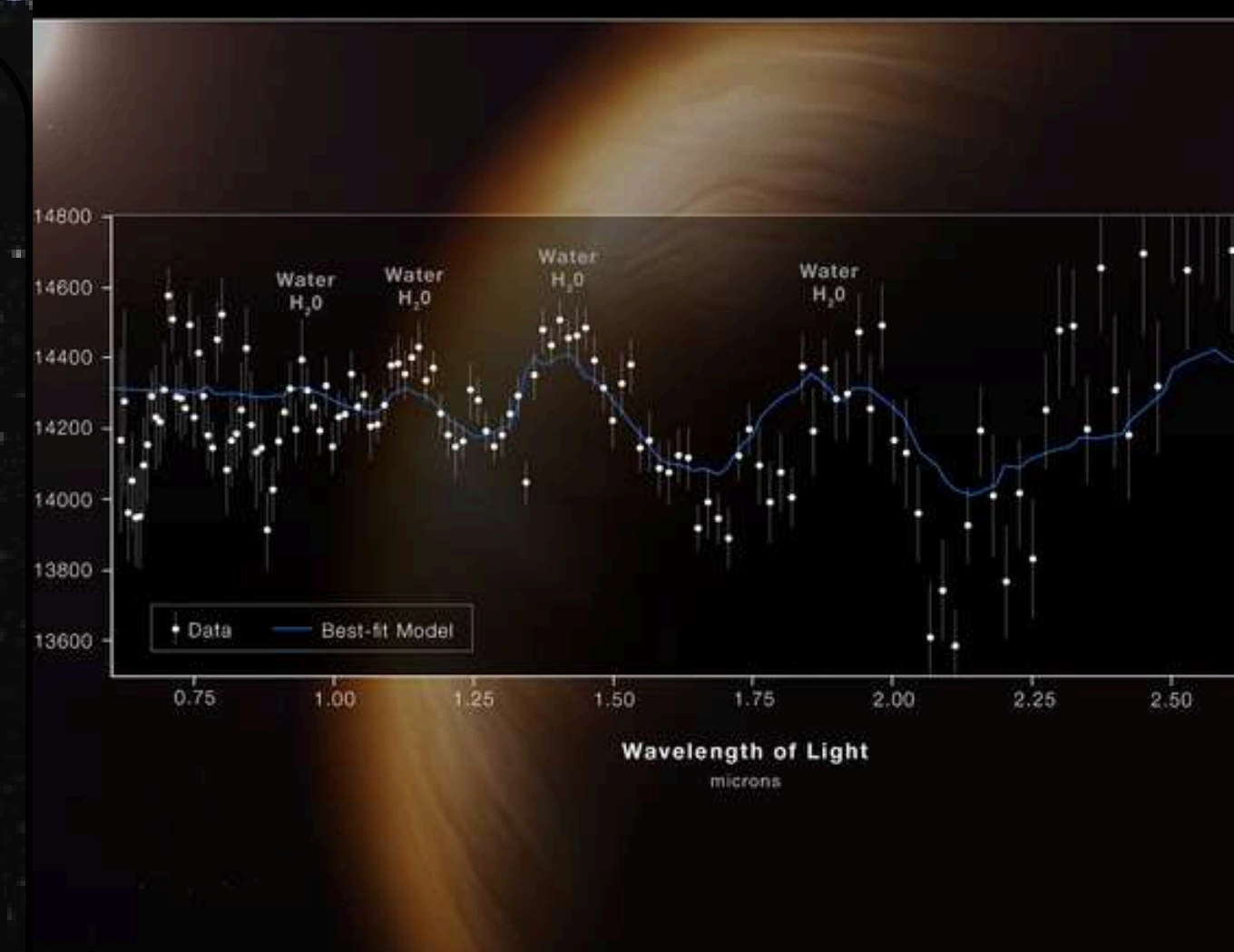
```
import matplotlib.pyplot as plt
plt.rcParams['figure.figsize'] = [10, 10]
plt.imshow(image, cmap='gray', origin='lower')
plt.show()
```



Spectroscopy studies how matter interacts with light, using each gas's unique spectral fingerprint to detect and measure atmospheric gases like CO₂, O₃, and NO₂.

Measurement techniques and instrumentation

- **FTIR Spectroscopy** : Captures infrared spectra from space to detect gases like CO, CH₄, O₃, and NH₃ and provides high resolution vertical profiles
- **LIBS** : Generates plasma of aerosols and analyzes emitted light for elemental composition. It also provides an *insitu* insight.
- **AAS** : It is used to quantify metallic atoms in atmospheric samples. Measures absorption of light by free metallic elements like lead.
- **AES** : Analyzes light emitted by excited atoms, often after the excitement by plasma.
- **Remote Sensing Spectroscopy** : Satellite-borne instruments measure the absorption or emission of atmospheric constituents over large spatial scales.



SPECTROSCOPY ANALYSIS USING PYTHON

- Data taken for analysis is derived from the PEPSI Exoplanet Transit Survey (PETS) and PEPSI Investigation, Retrieval, and Atlas of Numerous Giant Atmospheres (PIRANGA) projects.
- The captured spectrum is divided into 2 parts, blue (4800-5541 Å) and red (6728-7419 Å) arms of the spectrograph which we stitch using data analysis together to make a final overall spectra.
- We sort the combined spectra by ascending wavelength so the spectrum is continuous.
- Optionally, in any overlapping region, we average the flux values at identical wavelengths (or you could interpolate one onto the other's grid) to smooth the join.
- Then we invert the flux (i.e. subtract from its maximum) so that absorption dips become peaks.

```
from astropy.io import fits
import matplotlib.pyplot as plt
import numpy as np
from scipy.signal import find_peaks

blue_path = "/content/pepsib.20210501.009.dxt.nor"
red_path = "/content/pepsir.20210501.009.dxt.nor"

hdul_blue = fits.open(blue_path)
hdul_red = fits.open(red_path)

spec_blue = hdul_blue[1].data
spec_red = hdul_red[1].data

wl_blue = spec_blue['Arg']
fl_blue = spec_blue['Fun']

wl_red = spec_red['Arg']
fl_red = spec_red['Fun']

wl_all = np.concatenate([wl_blue, wl_red])
fl_all = np.concatenate([fl_blue, fl_red])

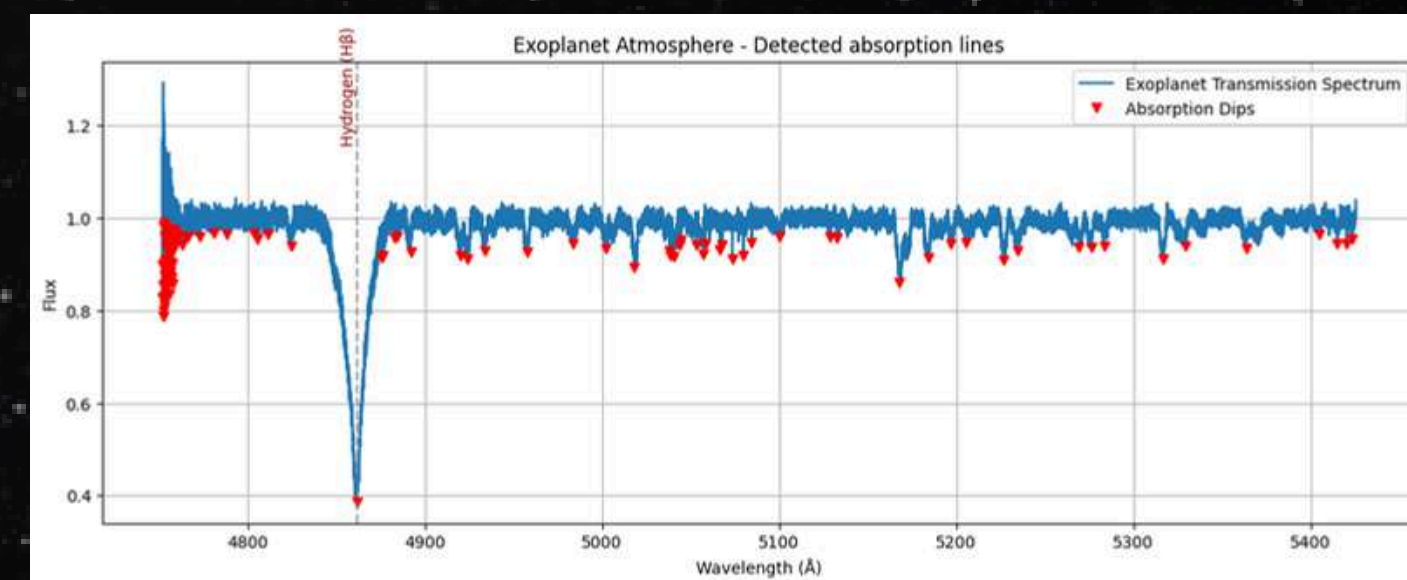
sort_idx = np.argsort(wl_all)
wl_comb = wl_all[sort_idx]
fl_comb = fl_all[sort_idx]

inverted_flux = np.max(fl_comb) - fl_comb
plt.figure(figsize=(12,5))
plt.plot(wl_comb, inverted_flux, lw=1)
plt.xlabel("Wavelength (Å)")
plt.ylabel("Inverted Flux / Intensity")
plt.title("Combined Exoplanet Spectrum (Blue + Red)")
plt.grid(True)
plt.tight_layout()
plt.show()

peaks, _ = find_peaks(inverted_flux, height=np.mean(flux)
peak_wavelengths = wavelength[peaks]
```

SPECTROSCOPY ANALYSIS USING PYTHON

- `scipy.signal.find_peaks` is designed to run on the inverted spectrum, with tuning parameters such as prominence, width, and distance, to pick out genuine atmospheric absorption lines, ignore noise and store the detected peaks in an array.
- A dictionary of known atomic or molecular peaks (from NIST Atomic Spectra Database) is imported into the code through which the detected peaks are checked against it.
- For each detected dip, we check if its wavelength lies within a chosen tolerance ($\pm$ a few Å) of one of these known lines from the dictionary, flagging likely matches.
- Additionally, the whole process could be automated by changing the prominence levels and the threshold values for the baseline helping in the smooth detection of elemental peaks.
- Furthermore, the value of each elemental abundance can be found out using the amount of flux for each element against the normal baseline flux using integration of equivalent width and conversion to column density.



The first part of the assignment was about determining distance of beehive cluster situated in cancer using different computational methods.

SOME THEORETICAL BACKGROUND

- **Apparent magnitude** – Apparent magnitude is a measure of how bright an object appears from Earth.
here F is the flux from the star, and F_0 is a reference flux

$$m = -2.5 \log_{10} \left(\frac{F}{F_0} \right),$$

- **Flux - luminosity relation** – By inverse square law

$$F = \frac{L}{4\pi d^2},$$

- **Distance modulus** – Linking apparent (m) and absolute (M) magnitudes here d_{pc} is distance in parsecs.
The Sun's absolute magnitude is $M_{\odot} = +4.83$

$$m - M = 5 \log_{10}(d_{pc}) - 5,$$

Approach 1:

- Exclude stars with $p_i < p_{min}$ (e.g. 0.8).
- Compute each star's absolute magnitude:
- Solve distance modulus for d_i (parsecs).
- Cluster distance: arithmetic mean of distance

Approach 2:

- Compute each star's absolute magnitude:
- Solve distance modulus for d_i (parsecs).
- Cluster distance: weighted mean of d_i with corresponding probabilities as weights

```
df = pd.read_csv('Beehive_data.csv')
df['prob'] = df['prob'] / 100
df['M'] = 4.83 - 2.5*df['logL']
df['dist'] = 10**((df['# Vmag'] - df['M'] + 5) / 5)
df['wdist'] = df['dist']*df['prob']
ls = [df['dist'][i] for i in range(len(df['dist'])) if df['prob'][i] >= po]
print(f"Mean distance of more certain stars: {sum(ls)/len(ls)} parsecs")
print(f"Weighted Mean distance : {(df['wdist'].sum())/(df['prob'].sum())} parsecs")
```

Python Implementation

The second part of the assignment involved analysis of gravitational waves, and data visualisation using matplotlib

ANALYSIS TASKS:

- Plot 1: Strain $h(t)$ vs. t
- Plot 2: Frequency f_i vs. t . Note the step-like behavior due to zero-crossing segmentation.
- Zero-Crossings: Identify indices where $h(t)$ changes sign. Use these to segment data.
- Scatter Plot: For each segment, plot (t_{zero}, f_i) to visualize frequency evolution.
- Optionally, fit a power law $f(t) \sim (t_c - t)^\alpha$ to the inspiral portion to estimate $\alpha \approx -3/8$.

```
data, freq, time = np.loadtxt('GW_data_file (1).csv', delimiter=',')
plt.plot(time, data)
plt.show()
plt.plot(time, freq)
plt.show()
eps = 5 #chose arbitrarily by seeing the frequency data
last = freq[0]
ztime = []
zfreq = []
for i in range(len(freq)):
    if abs(freq[i] - last) >= eps:
        last = freq[i]
        ztime.append(time[i])
        zfreq.append(freq[i])
plt.scatter(ztime, zfreq)
plt.show()
```

Python implementation

Machine learning is the ability of systems to learn from data, identify patterns, and make decisions with minimal human intervention, enabling automation and continuous improvement over time.

**HOW DOES
MACHINE
LEARNING
WORK? (THE ML
PIPELINE)**

- 1. Data Collection: Gathering relevant data.**
- 2. Data Preparation: Cleaning, preprocessing, and transforming data (e.g., handling missing values, scaling).**
- 3. Model Selection: Choosing an appropriate ML algorithm.**
- 4. Model Training: Feeding the prepared data to the algorithm to learn patterns.**
- 5. Model Evaluation: Assessing the model's performance on unseen data.**

CNN ARCHITECTURE

WHAT IS CNN MODEL?

Convolution Neural Network is a specialized type of neural network primarily used for processing grid-like data, such as images. It has 6 layers: Convolutional Layer, Activation Function (ReLU), Pooling Layer (Max Pooling), Flatten Layer, Fully Connected (Dense) Layers, Output Layer

Key Components

Conv2D (Convolutional Layer)

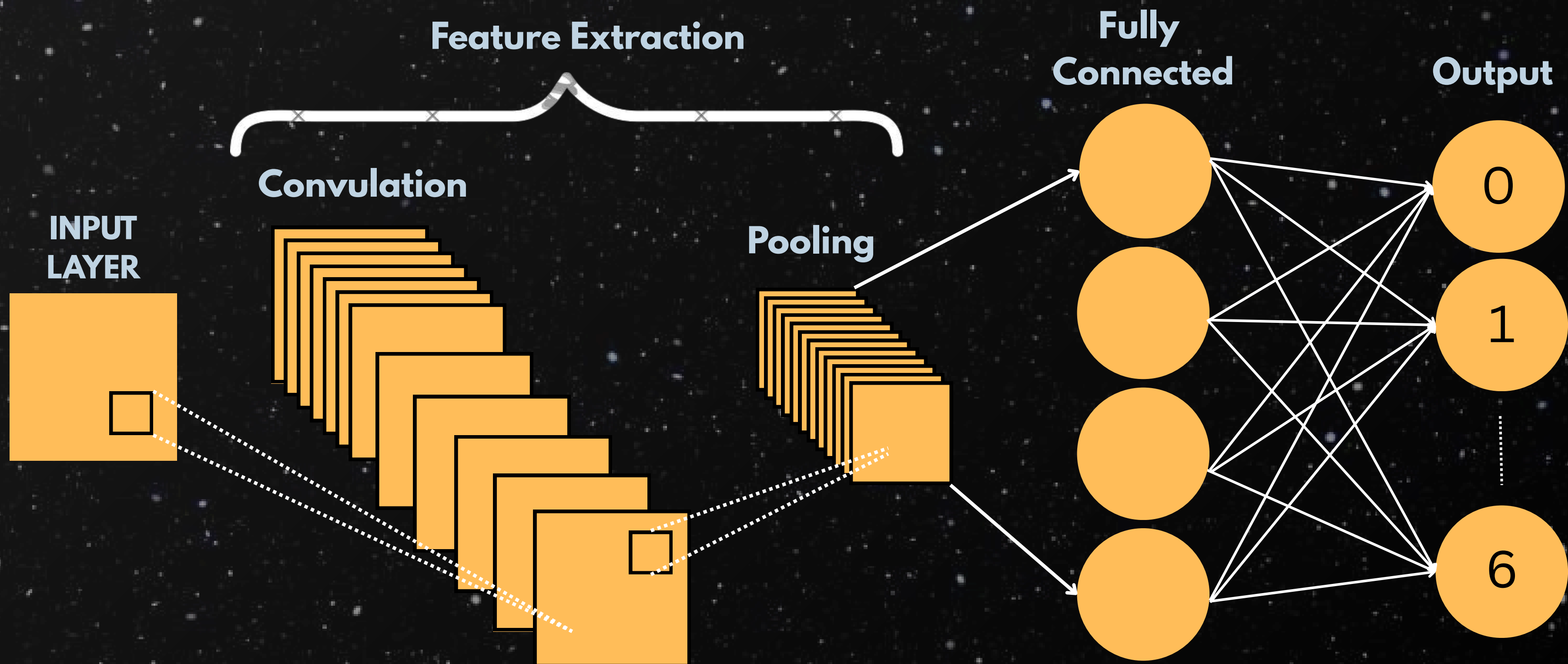
Dropout

MaxPooling2D

Flatten

BatchNormalization

Dense Layers



CNN is defined using Keras' Sequential() API:

Flatten

```
from tensorflow.keras import layers, models
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
from sklearn.utils.class_weight import compute_class_weight

model = models.Sequential([
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(64, 64, 1)),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dropout(0.5),
    layers.Dense(7, activation='softmax')
])
```


DATA ANALYSIS

Data analysis is essential step in building ML model as it helps us understand general characteristics of the data, analyse the quality of data, identifying any patterns, and prepare the data for modeling.

STEPS INVOLVED IN DATA ANALYSIS:

Summary of data

Getting general statistics of data, mean, median, standard deviation, etc.

Data visualisation

Visualising numerical data using histograms, box plots, scatter plots, etc.

Feature relationships

Analysing any correlation between different labels to identify any patterns

Handling missing values

Removing missing values or replacing them with the mean to improve model performance

Data preprocessing involves preparing the data in a format suitable for training the model, ensuring consistency, handling missing values, and transforming features to improve learning efficiency and accuracy.

Why is it needed?

To handle inconsistencies

To improve model performance and accuracy

To prevent bias

Steps involved

Handling missing values

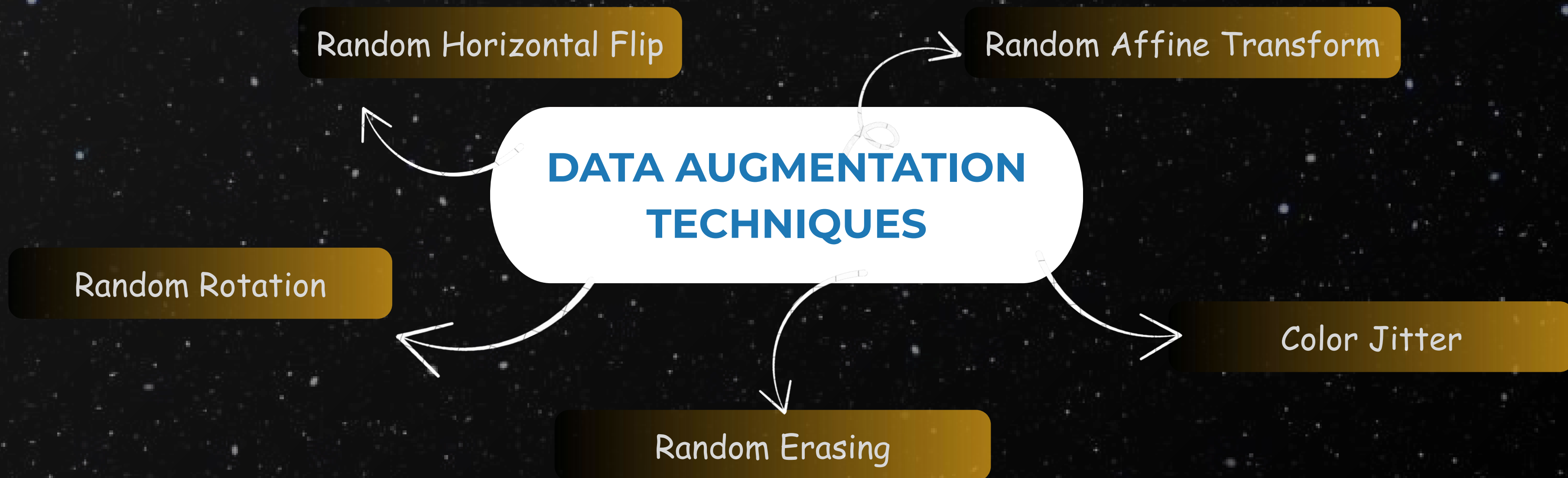
Removing outliers

Encoding categorical variables

Normalization and standardization

DATA AUGMENTATION

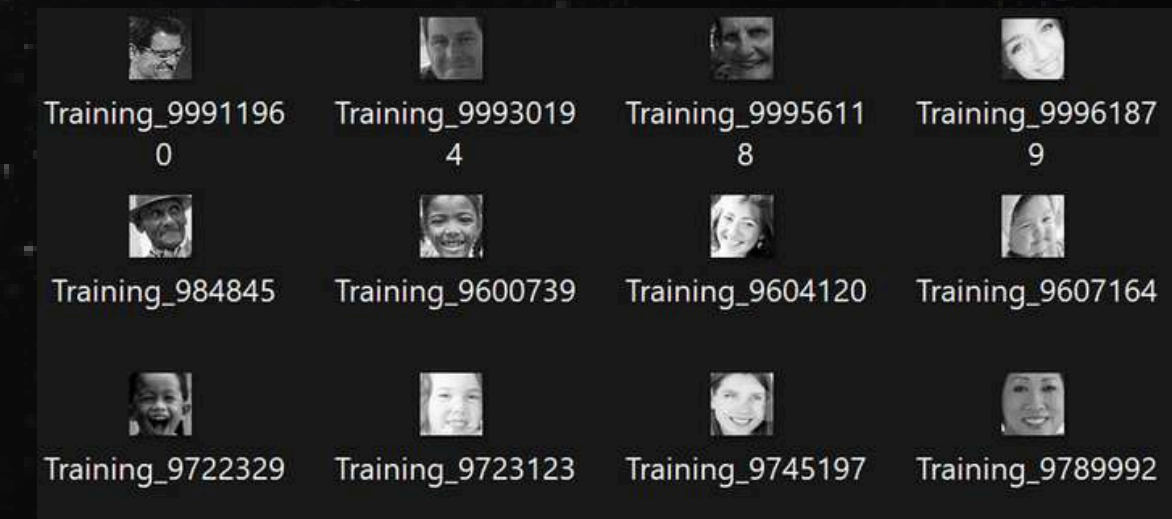
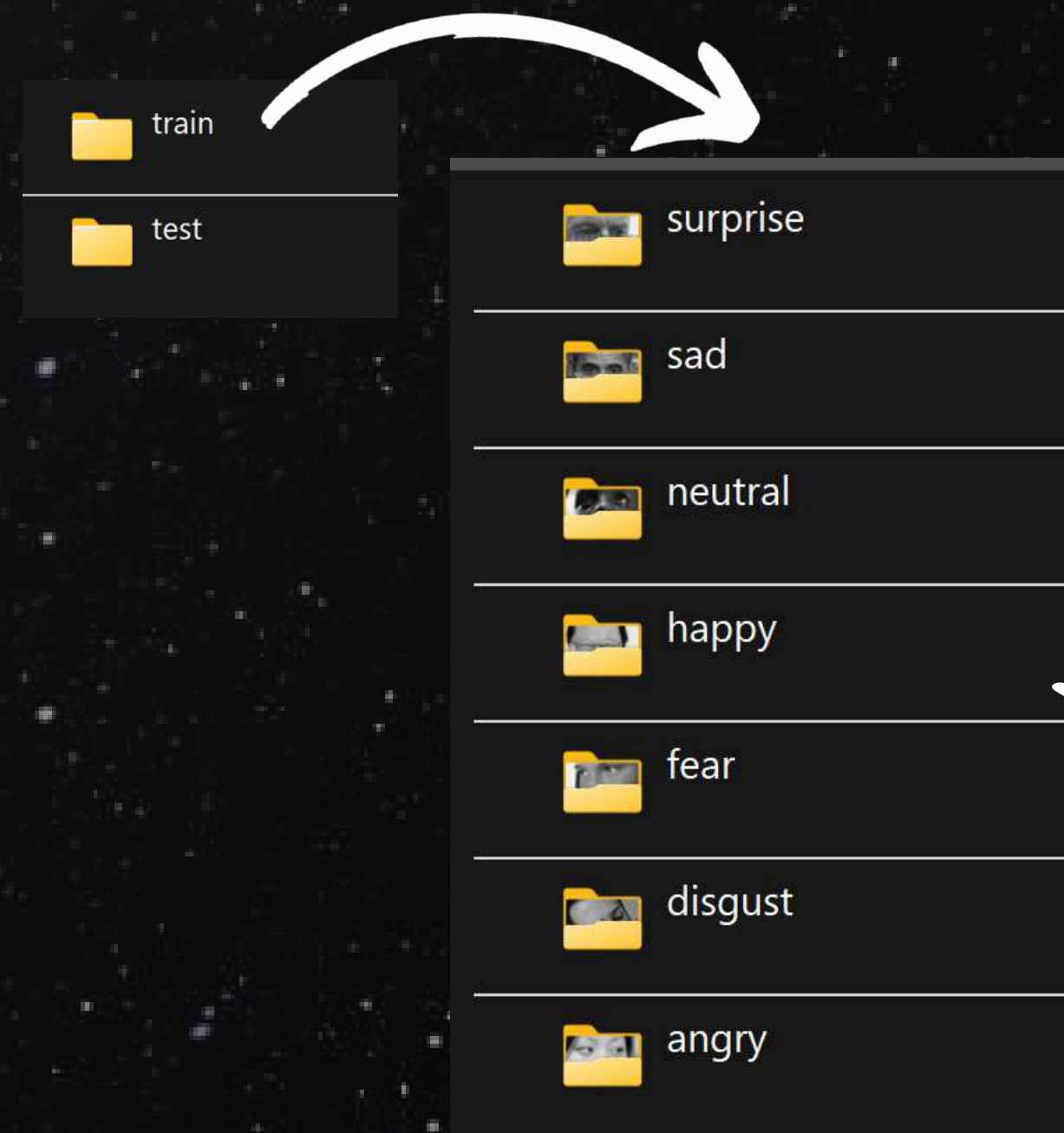
Data augmentation involves adding variations to the data in order to increase the training dataset, improve model performance, and enhance the model's ability to generalize to unseen data.



Build a CNN model for detection of facial emotion of a person for the dataset given:

EMOTIONS CLASSES

'Angry'
'Disgust',
'Fear'
'Happy'
'Sad'
'Surprise'
'Neutral'



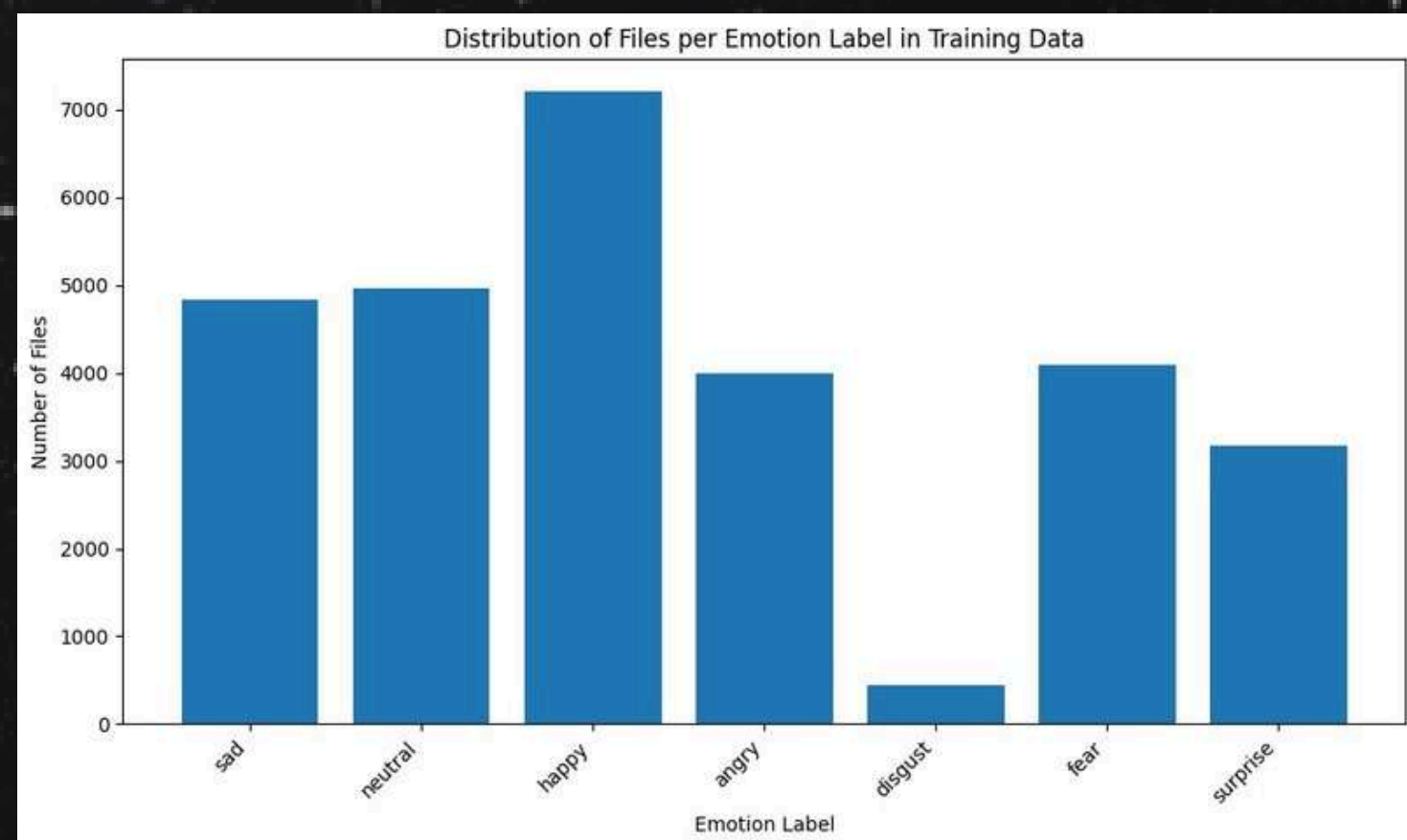
STEPS:

DATA
ANALYSIS

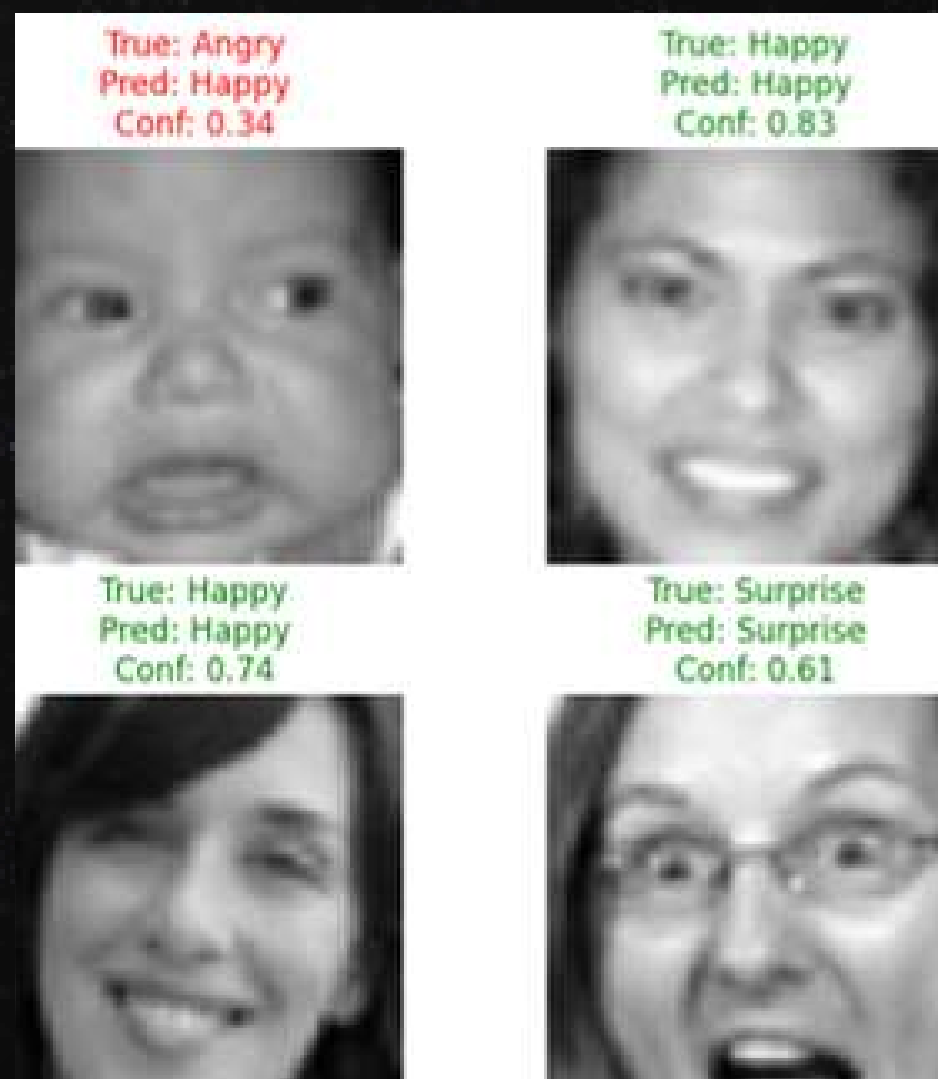
DATA
PREPROCESSING
AND
AUGMENTATION

MODEL
ARCHITECTURE
AND TRAINING

ACCURACY
TRAINING



B&W 48*48 PIXEL IMAGES



Classification Report:

	precision	recall	f1-score
Angry	0.39	0.34	0.36
Disgust	0.34	0.65	0.44
Fear	0.32	0.19	0.24
Happy	0.74	0.76	0.75
Sad	0.22	0.46	0.29
Surprise	0.65	0.63	0.64
Neutral	0.50	0.32	0.39
accuracy			0.49
macro avg	0.45	0.48	0.45
weighted avg	0.52	0.49	0.49

Project Objective & Data

Our goal was to estimate the Earth Similarity Index (ESI) for exoplanets using machine learning. ESI is a value from 0 to 1 indicating a planet's similarity to Earth.

Datasets Used

- Training Data: hwc.csv (labeled ESI values)
- Prediction Data: kepler.csv, TESS.csv

Key Features

9 planetary & stellar features were considered, including:

- P RADIUS, P PERIOD, P TEMP EQUIL
- S TEMPERATURE, S RADIUS, S MASS

Data Preprocessing

- Missing values imputed using mean.
- All features standardized with StandardScaler.

Modeling Techniques & Results

Two machine learning models were used to predict ESI:

Random Forest Regressor

R2 Score : 0.995082

MSE : 0.00008483417

Support Vector Regressor (SVR)

R2 Score: 0.89

MSE: 0.0039

Reasonable performance, but Random Forest was preferred due to superior accuracy and interpretability.

P TEMP EQUIL	High
P RADIUS	High
P ECCENTRICITY	Medium
P PERIOD	Medium

Top Earth-like Exoplanets Identified

The Random Forest model was applied to Kepler and TESS datasets to identify top Earth-like candidates.

Kepler Dataset

1	Kepler-296 e	0.922824
2	Kepler-1512 b	0.892083
3	Kepler-442 b	0.852161

TESS Dataset

1	TRAPPIST-1 e	0.892674
2	TRAPPIST-1 d	0.857618
3	LHS 1140 b	0.850860

THANK YOU

